

LMS Seminar

On Exotic Materials in 3D Linear Elasticity with High Symmetry Classes

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Date, time, and venue

September 03, 2026 (2 – 3 pm), Amphithéâtre Pole Mécanique

Abstract

An anisotropic elastic material is referred to as exotic when, under specific loadings, its mechanical response exhibits a higher degree of symmetry than that prescribed by its intrinsic material symmetry. Such materials, which may be regarded as lying conceptually and functionally between two distinct symmetry classes, are of significant practical relevance. They enable the tailored design of metamaterials capable of reconciling otherwise incompatible mechanical requirements — for example, achieving directional isotropy of the Young's modulus in an intrinsically anisotropic medium. This work focuses on the systematic classification of exotic structures within the framework of three-dimensional linear elasticity. An exhaustive classification is carried out, leading to the enumeration of 18 exotic structures corresponding to symmetry classes higher than orthotropy. Representative examples of exotic elastic behaviours are analysed in detail.

About the speaker

Nicolas Auffray has been a Professor at Sorbonne University since 2022, where his research focuses on the study and design of architected materials. The notion of symmetry is central to his approach: his work addresses mathematical homogenization and the description of anisotropy, in connection with classical and generalized continuum mechanics. A recipient of the Jean Mandel Prize in 2015, he currently serves as co-director of the GDR Archi-Meta and has taken part in various ANR projects (ArchiMatHOS, Max-Oasis, MoMaP...) on these topics.

